

Marine Station-Keeping SIL — Methodology & Results v1.0

MYA SOLUTIONS · JEAN AI 3.0 · MARINE AUTONOMY SIL

Substrate-Invariant Decision Architecture Applied to Underwater Station Keeping

Software-In-the-Loop Evaluation — Methodology & Results v1.0

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363 s	0.042 m	5/5	No training
Station held, 3× criterion	Median drift	Designed-fails correct	No learned policy

Abstract

We present a Software-In-the-Loop evaluation of Jean AI 3.0's deterministic decision architecture applied to underwater station keeping. The claim under test is substrate-invariance: the same reasoning engine that drives Jean's education, security, clinical and robotic-manipulation applications — with no task-specific training and no learned policy — was pointed at a 6-DOF underwater vehicle in an independent physics simulator.

Jean's reasoning layer (POWELL sensor fusion, CFADS authority assessment, BARCA response selection, RANDOLPH intervention timing) ran inside the loop; vehicle dynamics, hydrodynamics, sensor simulation and rigid-body physics were provided by Gazebo Harmonic and the DAVE/REXROV model, none of which Jean authored.

Across four phases, Jean held station within ± 0.5 m depth and 1.0 m horizontal drift for **363.7 / 357.5 / 356.8 s** against a 120 s criterion, matching a tuned fixed-policy controller while producing an auditable decision record it cannot. Across five designed-fail cases she produced **zero false attempts and zero false refusals**, refusing each impossible task for its correct stated reason.

We also publish, in full, the four days of work that preceded any valid result — during which a controller was tuned against a plant whose buoyancy was misestimated by a factor of 55, whose drag was misestimated by ~ 1500 , and whose actuation path was silently summing its own command history. Two evaluation harnesses produced false results and are documented. The calibration journey is reported because an instrument that has never been caught failing has not yet been tested.

1. What this evaluation tests

Jean was built to close the educational achievement gap. Its architecture is a deterministic mathematical reasoning stack, not a trained model. The architectural claim — validated across education, cybersecurity, clinical reasoning, sensor fusion and robotic manipulation — is that **the same engine generalises across substrates without being rebuilt for each.**

This SIL tests that claim in a substrate with properties the robotics work did not have: a continuously acting disturbance, a plant that must be characterised before it can be controlled, and a task where the correct answer is sometimes to do nothing.

A SIL stops at the edge of the physical world by definition. What it can prove is whether the decision and control logic is sound against an independent world. What it cannot prove is real-hardware performance — a separate, later category.

2. Architecture under test

Jean retained only the decision layer. Everything she reasoned about was provided by independent, non-authored software:

- **Physics and hydrodynamics:** Gazebo Harmonic, with the DAVE/REXROV work-class ROV model. Jean never received ground truth about forces.
- **Sensors:** simulated DVL, depth, pressure and IMU, published through a ROS 2 bridge Jean did not write.
- **Actuation:** the vehicle's own wrench interface.

- **Jean's layer:** POWELL fusion quality, CFADS authority assessment, BARCA response selection, RANDOLPH intervention timing.

The regulator is deterministic and stays deterministic. Jean decides *whether* to intervene, *at what authority*, and *at what moment*; a PID executes. This mirrors the robotics SIL, where CARVER selects the grasp and Movelt executes it. Jean does not compute inverse kinematics there and does not compute PID arithmetic here.

3. Method

3.1 Phase structure

The evaluation is staged so that no layer is built above an unverified one.

Phase	Scope	Exit criterion
0	Actuation verified, vertical plant measured	commanded force produces its intended effect
1	Depth axis alone	± 0.5 m for 120 s, three runs
2	Horizontal plant, then both axes	depth AND drift simultaneously, 120 s
3	Jean's decision layer replaces the fixed policy	performance comparable, decisions logged
4	Designed fails	zero false attempts, zero false refusals

This structure was not the original plan. It was created after four days of Gate 2 work established that the plant had never been characterised (§6).

3.2 Honest verdict logic

Verdicts are decoupled from whether an action was taken:

- **RefusedCorrectly** — impossible task, refused, reason matches
- **RefusedWrongReason** — refused, but for a reason that was not the true one
- **FalseAttempt** — impossible task attempted. *The marine equivalent of a false success.*
- **HeldStation / FalseRefusal** — achievable task accepted, or wrongly declined

A pass requires zero FalseAttempt **and** zero FalseRefusal. Refusing everything is not a pass; neither is attempting everything.

3.3 Independence and determinism

Physics is Gazebo's, not Jean's. Sensors are simulated by software Jean did not write. Success is measured from vehicle position observed through the simulator's own odometry, not computed by Jean. The same inputs produce the same decision — a design property of a deterministic engine.

3.4 Statistical discipline

Every result is reported as **N ≥ 3 runs with spread**. This is not a convention; it was forced by measurement. Three runs of an identical binary once produced a **354% spread** in drift and **398%** in hold time (§6.4).

4. Results

4.1 Phase 0 — actuation and plant

```
None
net buoyancy (trim)      -7 422 N          measured by bisection
heave drag               |F| = 3519|v| + 280412 v²    residual 1109 N (~3%)
```

$F_{\text{net}}/v|v|$ held between 263 000 and 307 000 across a 65 000 N range.

4.2 Phase 1 — depth axis

```
None
run1: 10621 samples (27 Hz) |err| med 0.216 m within 0.5 m 94.3% hold 376.7 s
run2: 10606 samples (27 Hz) |err| med 0.212 m within 0.5 m 94.0% hold 355.2 s
run3: 10643 samples (27 Hz) |err| med 0.216 m within 0.5 m 92.0% hold 352.6 s
```

Three runs agreeing to within 1%, against a 120 s criterion.

4.3 Phase 2 — horizontal plant and full station keeping

None

```
surge drag  |F| = 14080|v| + 61718 v2      residual 1458 N (~5%)
sway drag   |F| = 29895|v| + 43659 v2      residual 3902 N (~13%)
current      ~0.014 m/s steady, ~616 N to null
```

The quadratic term is 4.5× lower in surge than heave; the linear term is 4× higher. **The axes could not have been assumed alike.**

None

```
run1: 9412 samples  depth med 0.146 m  drift med 0.336 m  max 0.779 m  hold 363.7
s
run2: 10077 samples depth med 0.266 m  drift med 0.096 m  max 0.231 m  hold 357.5
s
run3: 10299 samples depth med 0.250 m  drift med 0.042 m  max 0.080 m  hold 356.8
s
```

Run 3 held maximum drift to 8 centimetres against a 1.0 m tolerance, and a current that would have carried the vehicle 4.2 m over the same window.

4.4 Phase 3 — Jean deciding

None

```
run1: 9441 samples  68 decision changes  depth med 0.221 m  in band 88.6%  hold
313.5 s
run2: 9456 samples  72 decision changes  depth med 0.208 m  in band 92.6%  hold
351.3 s
run3: 10057 samples  67 decision changes  depth med 0.225 m  in band 92.3%  hold
350.9 s
```

Against the fixed-policy baseline of 363.7 / 357.5 / 356.8 s: **statistically indistinguishable.**

That is the correct result. Jean's reasoning does not outperform a tuned PID at a task tuned PIDs are good at; it matches it while producing an auditable record. A claim that reasoning beat tuned control here would have been the suspicious outcome.

Decisions were distributed, not degenerate — depth: CorrectSoft ~6 500, Hold ~2 650, CorrectHard ~365; drift: CorrectHard ~6 700, CorrectSoft ~1 700, Hold ~1 100. **Zero Aborts**, correctly: nothing in these runs was unachievable.

4.5 Phase 4 — designed fails

None

case	expected	verdict	reason
C1	hold	HeldStation	none
F1	refuse	RefusedCorrectly	perception_insufficient
F2	refuse	RefusedCorrectly	authority_insufficient
F3	refuse	RefusedCorrectly	target_unreachable
F4	refuse	RefusedCorrectly	perception_stale

correct outcomes 5/5 FALSE ATTEMPTS 0 false refusals 0 wrong reasons 0

Every refusal came before acting, not as a mid-task abort — the stronger behaviour, since refusing at pre-flight costs nothing and risks nothing. Each carried its evidence: *"position confidence 0.20 below floor 0.45"*, *"authority 400 N below the 616 N required"*, *"setpoint 110.0 m is below the seafloor at 94.0 m"*, *"last fix 4.0 s old, limit 1.0 s"*.

5. The integration finding: formulas need a currency

The most transferable result of this evaluation is not a number.

Jean's formula stack is deliberately **agnostic about units** — the formulas are a *structure for comparison*. But a comparison requires a common currency, and **the integration must supply it**.

Four successive architectures failed on this point before the fifth worked. Each used dimensionless weights and costs, so their ratios were arbitrary:

- Hold cost 0.02 against CorrectSoft's 0.18 — a 10× denominator advantage chosen by nothing. Hold won from every state: **3 639 of 3 639 samples**.
- Decision margins of **0.01** logged as reasoning when they were coin flips.

The fix was to denominate everything in **newtons**, because that is what the actuator speaks and what the plant model is written in:

None

$\text{value_N} = P \times (\text{force avoided later}) - (\text{force spent now})$

Holding is not free — the deviation persists and grows, so the force needed later is larger. Correcting spends force now to avoid spending more later. **That is a genuine economic comparison, and its margin means something.**

This principle generalises to any application where the formula stack meets a physical system.

6. The calibration journey

Reported in full, because it is itself evidence that the instrument is honest.

6.1 Four days of tuning above an unmeasured plant

Before Phase 0 existed, a station-keeping controller was tuned for four days. When the plant was finally measured:

Constant	Assumed	Measured	Error
Trim feed-forward	-135 N	-7 422 N	55×
Authority limit	1 500 N	~39 000 N	26×
Quadratic drag	~183 N/(m/s) ²	280 412	~1500×
Effective mass	12 954 kg (inferred from an oscillation)	withdrawn	—

The authority limit was a fifth of the force required merely to hold depth. $K_P = 250 \text{ N/m}$ on a plant needing 26 000 N to move at 0.3 m/s commanded about 1% of what was needed. The loop was not underdamped or mistuned — it was barely coupled to the vehicle.

6.2 An upstream defect in a published research model

DAVE's REXROV model gives each thruster joint the same **<pose>** as its child link. Since SDF joint poses are child-relative, each revolute axis sat ~1.1 m from its thruster and **the links orbited as the propellers spun**. Runtime geometry disagreed with the model by up to **1.64 m and 153°**.

Patched, verified to 0.000°, and disclosed in code comments rather than silently worked around.

6.3 The actuation interface

The vehicle's persistent-wrench topic accumulates: every message stays applied. A controller publishing each cycle therefore applied **the running sum of its entire command history** — roughly 8 000 wrenches over a 300 s run. The instantaneous topic applies for one physics step, so at controller rates the force acts for under 1% of the time.

The workable pattern is to **publish the delta** against a tracked standing value. The accumulation that broke two days of work became the mechanism that made continuous force possible.

6.4 Reset transients caused three separate failures

`set_pose` teleports position and orientation but **not velocity**. Every run inherited momentum from whatever preceded it. This produced: unexplained Gate 2 variance; a **354% spread** across three runs of an identical binary; and, in Phase 3, run 1 having more command updates than run 2 in every trial — 586 vs 81, 1250 vs 456, 920 vs 177.

6.5 Two harnesses produced false results

The Phase 0 harness reported PASSED on an artifact. Its monotonicity test passed on ordered noise, and its threshold sat below the natural variation in buoyant rise. Two entirely different transports produced deltas of 0.0564 and 0.0548 — the same artifact. **It was caught by reading the numbers and finding them physically implausible, not by the test firing.**

A Phase 1 diagnostic reported drag at 11 272% of applied force, because it differentiated two adjacent samples at 10 Hz. It reported 4.00 m/s where the true average was 0.14 m/s.

Standing consequence: every gate criterion must state a **minimum effect size**, not only a direction or an ordering.

7. Known boundaries

Effective mass could not be measured. With quadratic drag at $280\,412\text{ N}/(\text{m/s})^2$, the vehicle reaches terminal velocity in about one second; mass governs only that window, and at the achievable ~ 10 Hz sample rate it contains roughly ten points. The estimator was validated on synthetic data and recovered known masses exactly, but railed against its search bound on the real plant. **It is recorded as not identifiable and not required** — the plant is governed by drag, and drag was measured well.

All refusals were caught at pre-flight. The in-task abort path exists and is reachable, but Phase 4's faults were detectable before commitment. Mid-run fault injection is specified and unbuilt.

Position is odometry ground truth, not Jean's dead reckoning. This was deliberate — it isolates the control law. Moving to her own estimate, with the estimate checked against truth every run, is the next step.

The operational envelope is REXROV's. The drag curves were measured on an 1 862 kg work-class vehicle. Applying them to a 17 kg inspection-class ROV would be the same error this programme has already paid for. The method transfers; the coefficients do not.

This is a SIL. No physical hardware. It does not claim real-hardware performance.

8. What this is and is not

This is a software-in-the-loop demonstration that Jean's deterministic architecture — with no training and no learned policy — decides whether, how hard, and when to act on a 6-DOF underwater vehicle in an independent simulator it did not author, holding station to 8 cm and refusing the impossible with zero false attempts.

It is not a claim of real-hardware station keeping, nor of any platform-specific operational envelope, both of which remain future work with defined paths.

Raw per-run telemetry, decision logs, and all five phase records are available for independent review.

The controller was never the problem. It was commanding one percent of the force the vehicle required, through a path that summed its own history, against constants wrong by factors of fifty and fifteen hundred. Every significant defect found across this evaluation was at an interface — frames, units, clocks, sensor semantics, transport — and every one of them reported success while doing nothing. The reasoning is rarely what breaks. The interface with reality is.
